WQF7004
Data Analytics in Artificial Intelligence

Week 1

Introduction to Data Analytics

Learning Outcomes

- In this module, you will learn about data analytic concepts.

Topic Title	Topic Objective
Introduction	Understand what is data analytics and its importance
Type of data analytics	Understand and differentiate the four types of data analytics
Data scientists	State the role of data scientists

Contents

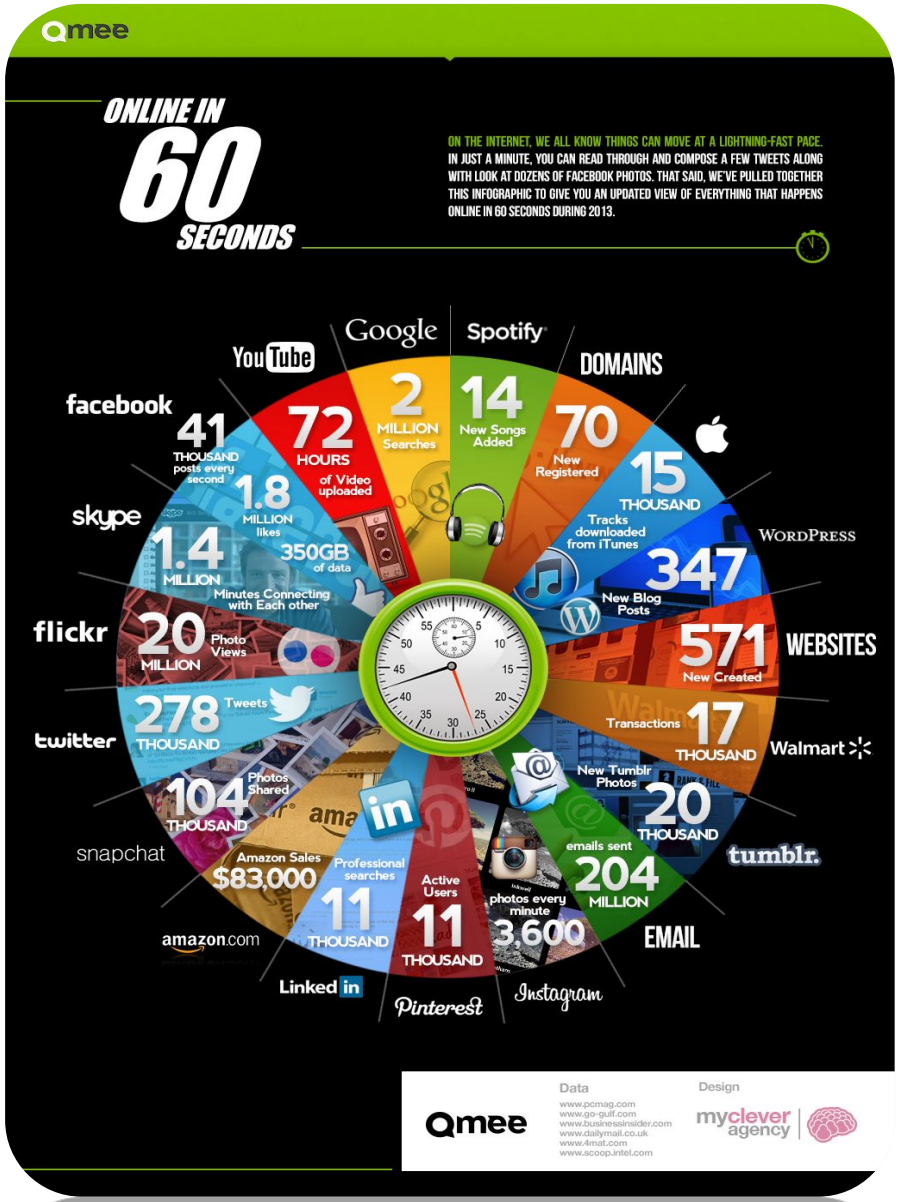
- Introduction
- Types of data analytics
- Importance of data analytics
- Role of data scientists
- Data analytics applications
- Summary
- Key Terms
- References

Introduction

- Data analytics is an emerging technique that dives into a data set without prior set of hypotheses.
- The role of Data Analyst/Data Scientist is closely related to Business Analyst, Business Intelligence, Statisticians.
- The data derive meaningful trends or intriguing findings that were not previously seen or empirically validated.
- Data analytics enables quick decisions or help change policies due to trends observed
- Accumulation of raw data captured from various sources (i.e. discussion boards, emails, exam logs, chat logs in e-learning systems) can be used to identify fruitful patterns and relationships) . If you are from the industry, your data would probably be some excel database, some pdf, text files, (or even images) of from some database system.

Data

- Data is created constantly, and at an ever-increasing rate.
- Substantially more challenging to analyzing vast amounts of the data.
- Big Data-Huge volumes of data of various types and sources.



Look Deep into Data

100%

13%

Core analytics

It is not about more data, but more in the quality of data and about the deeper look

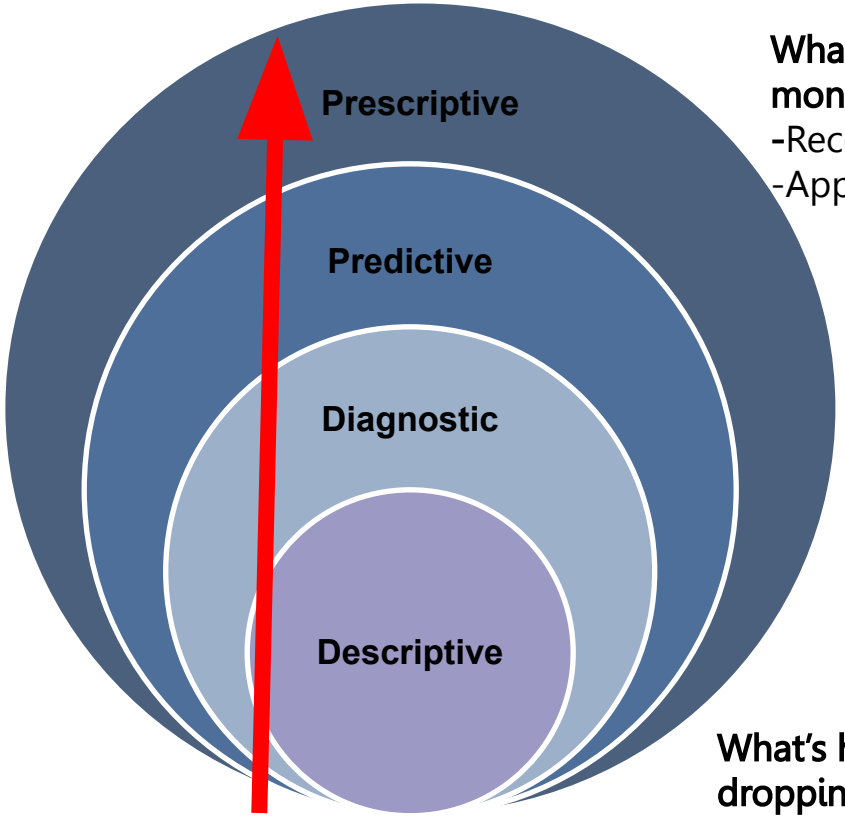
Advanced analytics

Analysis vs Analytics

- Analysis refers to the process of separating a whole problem into its parts so that the parts can be critically examined at the granular level.

- Analytics is a variety of methods, technologies and associated tools for creating new knowledge/insight to solve complex problems and make better and faster decisions

Type of Data Analytics



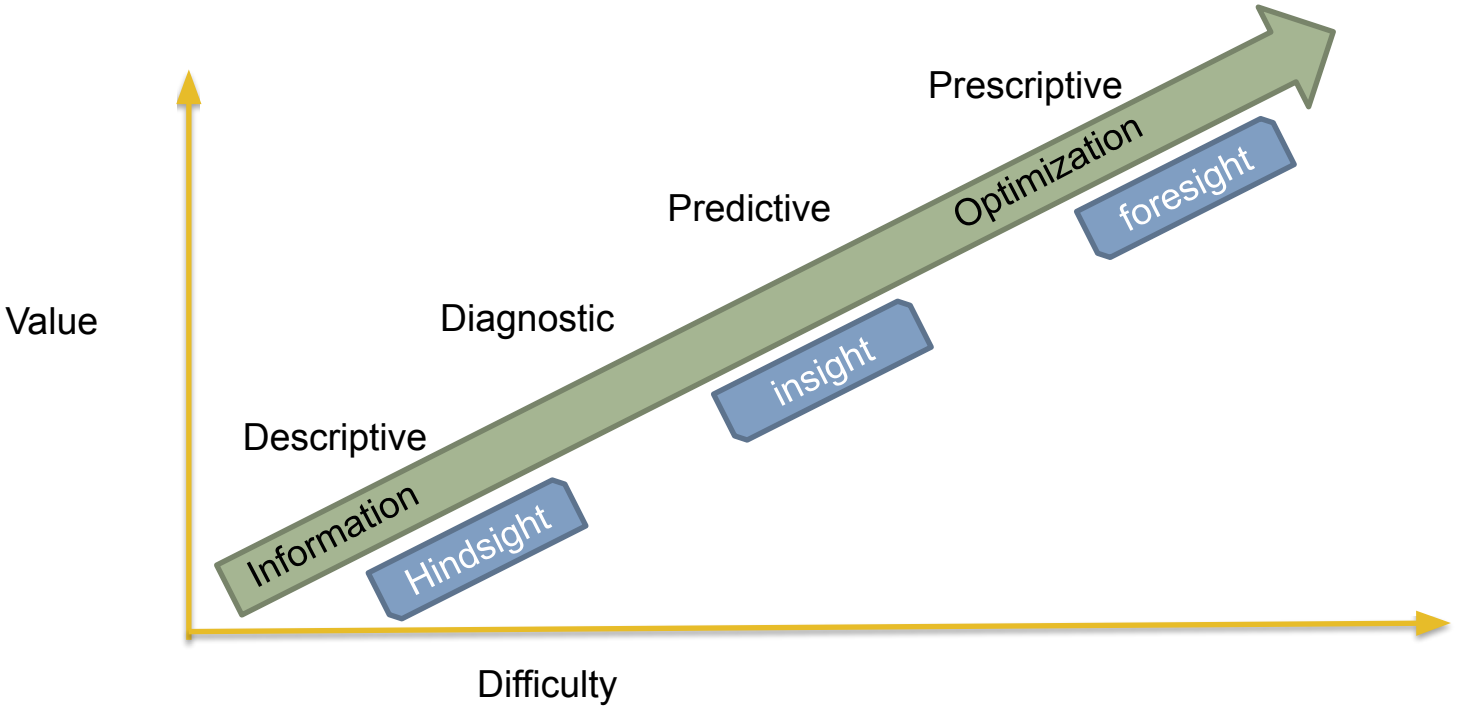
What do I need to do?- Plan strategies for the monsoon season
-Recommended actions and strategies
-Applying advanced analytical techniques

What's likely to happen?- Prediction-Machine Learning Comes in-Predict when is the next monsoon/when is the next time where sales are predicted to drop
-business strategies consistent over-time
-current/historical pattern predict the future
-Decision Automated

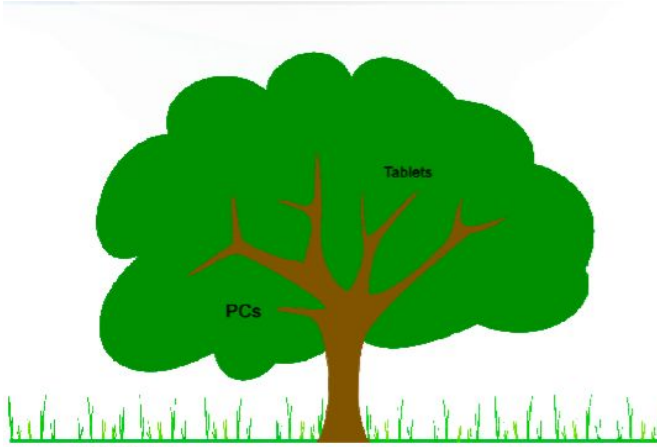
Why is it happening?-Because of the monsoon season.
-root-cause
-isolate confounding information

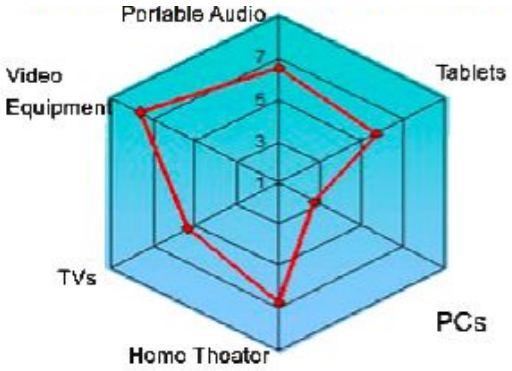
What's happening?-Sales are dropping
-comprehensive and live data

Type of Data Analytics



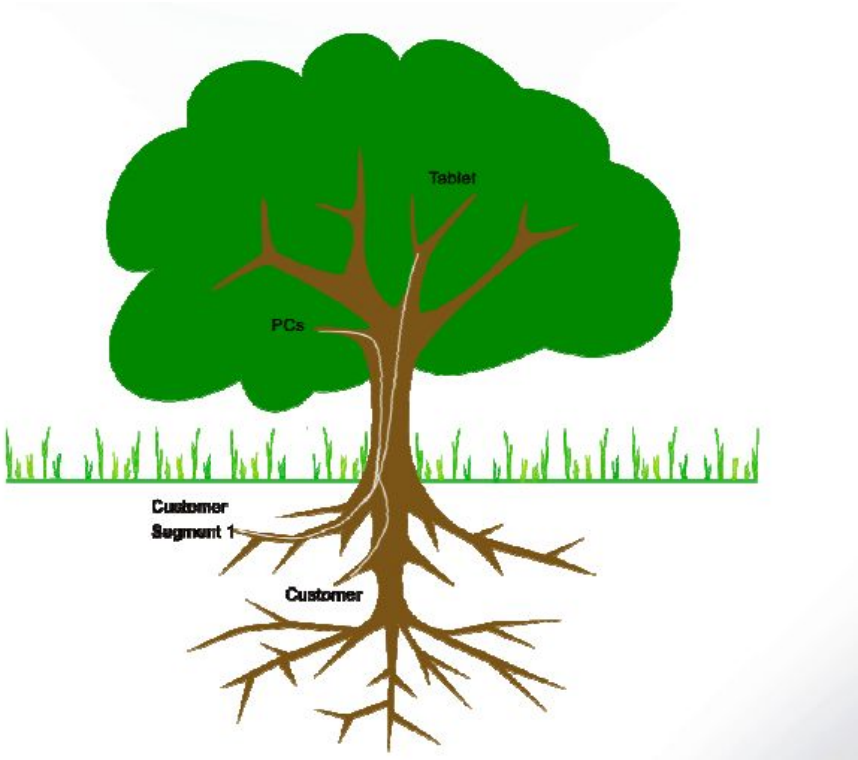
Descriptive Analysis - What happened?





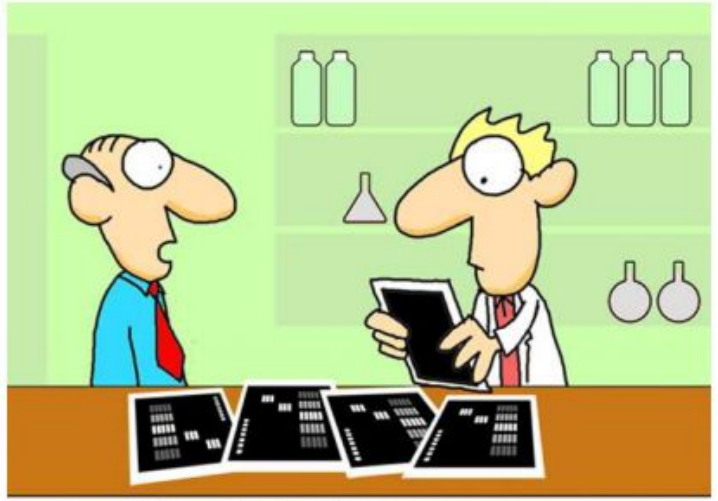
Total Sales:	$5,087
Operating Cost:	$5,590
Issues:	7
Processes Affected:	521

Diagnostic– Why is it happening?



- Look into the root / cause of certain problem

Predictive– What's likely to happen?



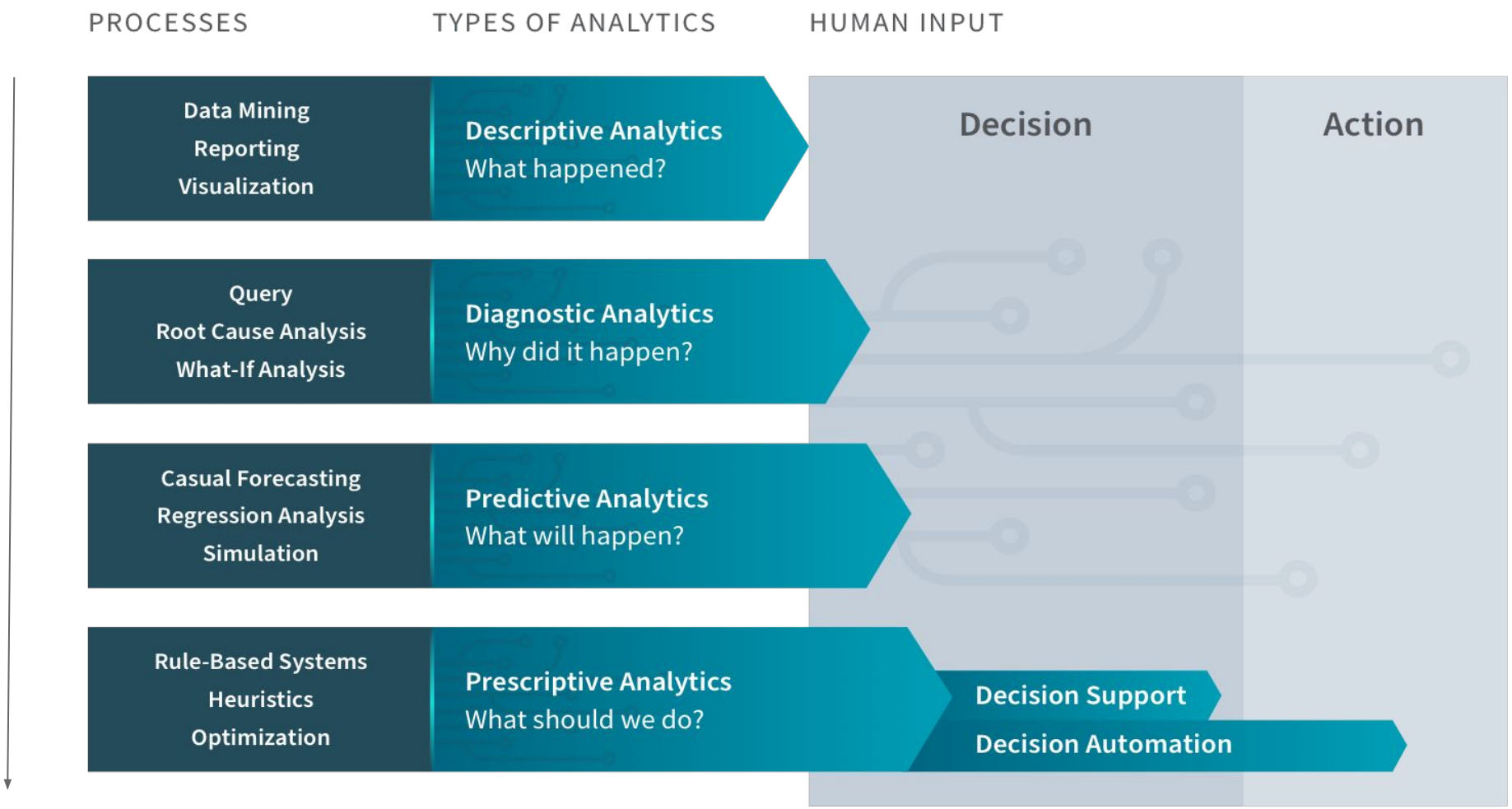
"Data don't make any sense,
we will have to resort to statistics."

- Bad statistics can cause the data look good!
- 50% of scientific articles use wrong methods!

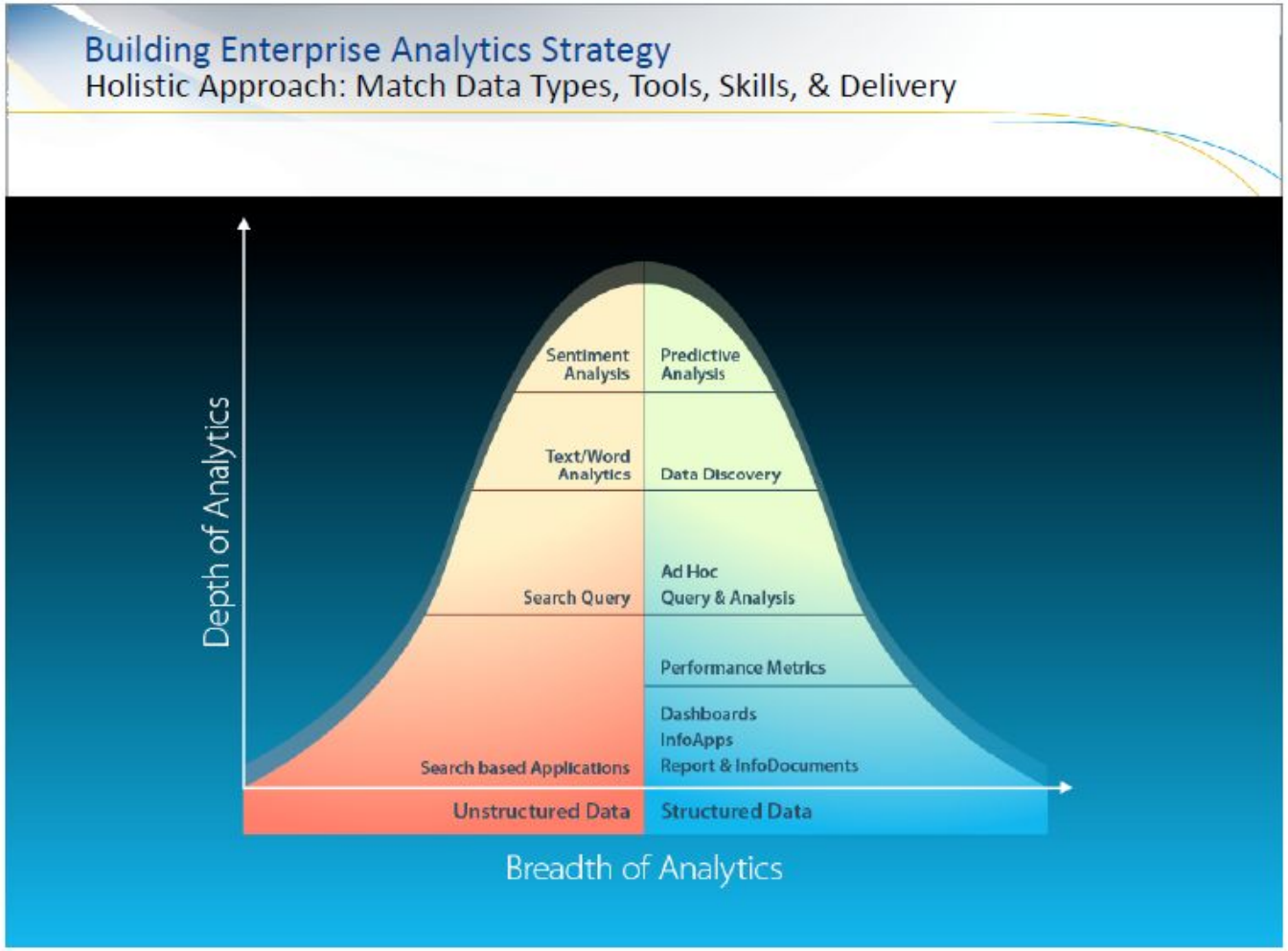
Prescriptive– What do I need to do?

Can we simulate various outcomes to choose the best?

Types of Data Analytics



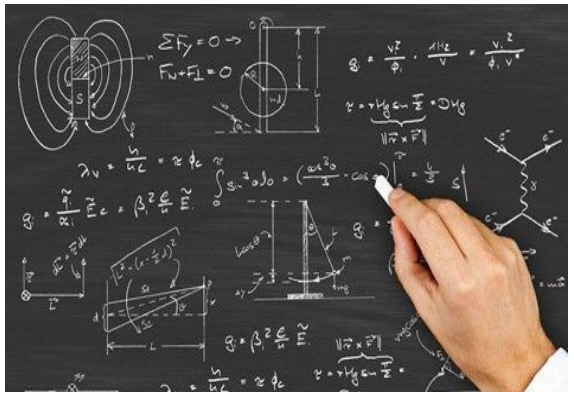
Types of Data Analytics



Data Scientist

What do people think Data Scientist can do?

Lots of data

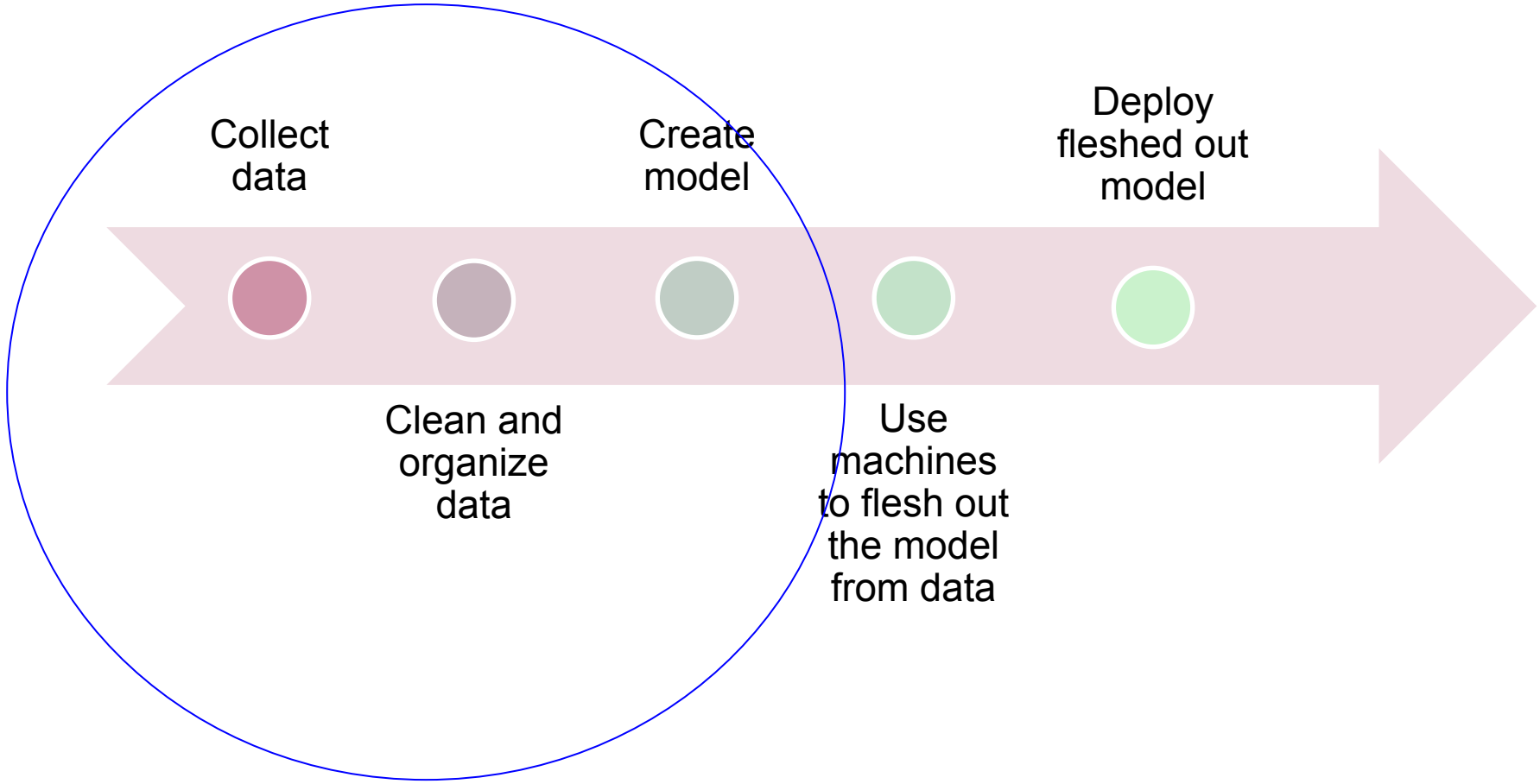


Complex mathematics in high-dimensional spaces

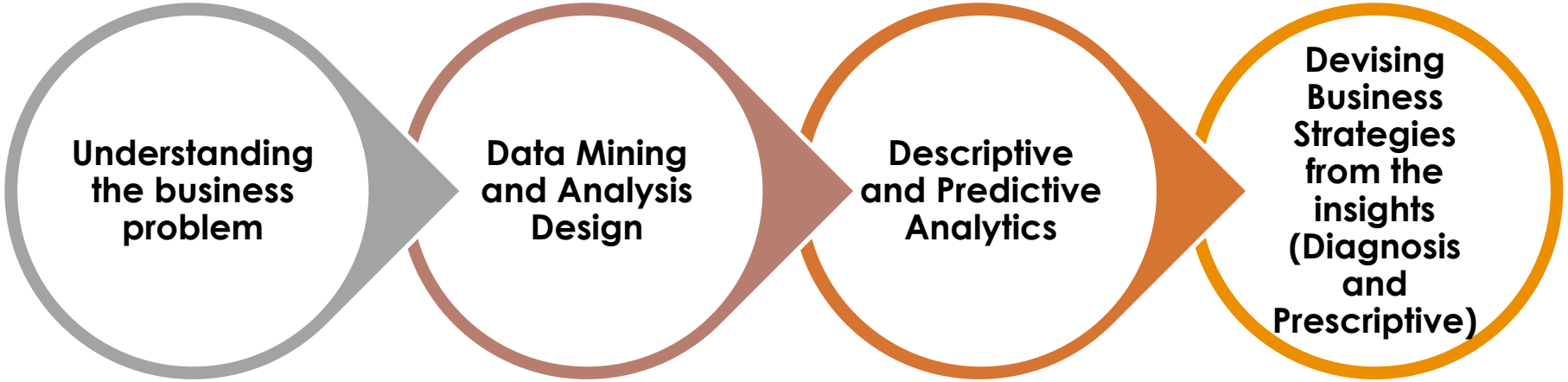
Magical results

Data Scientist

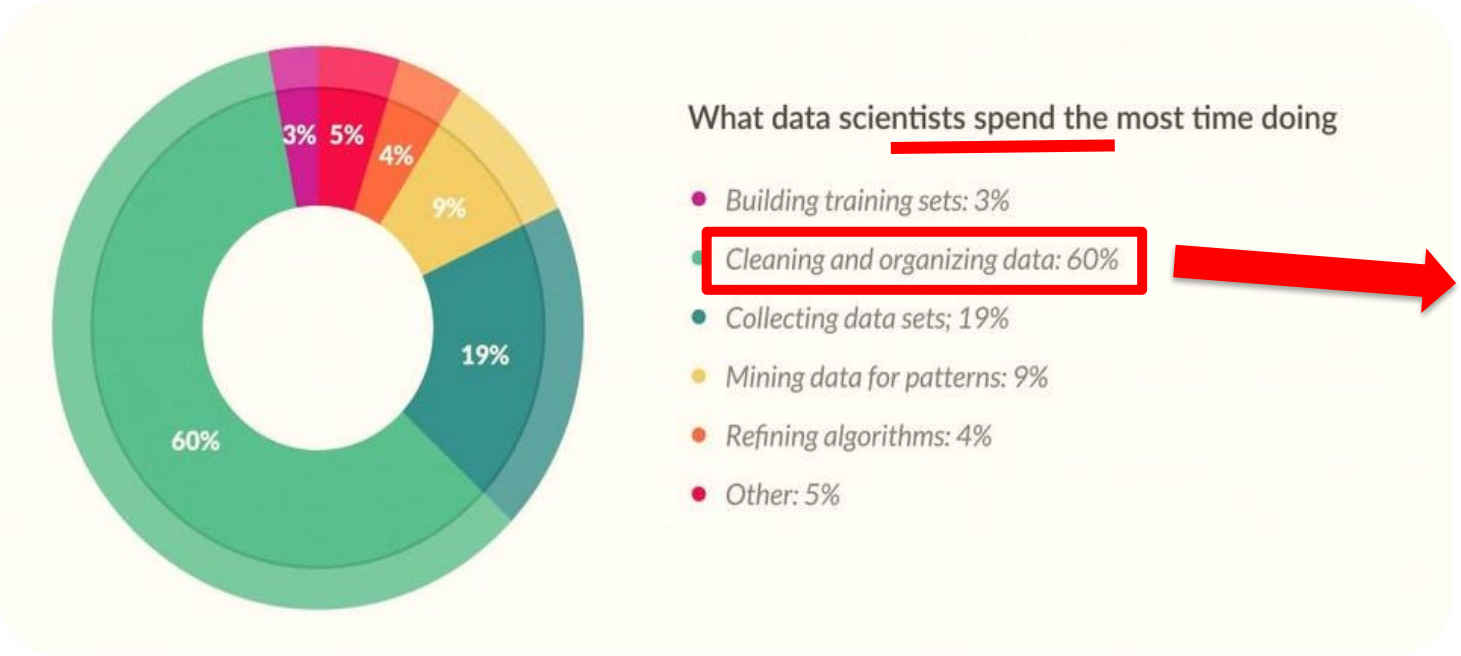
In reality,



Key functions of data scientist



Key functions of data scientist (more to Data Engineering)

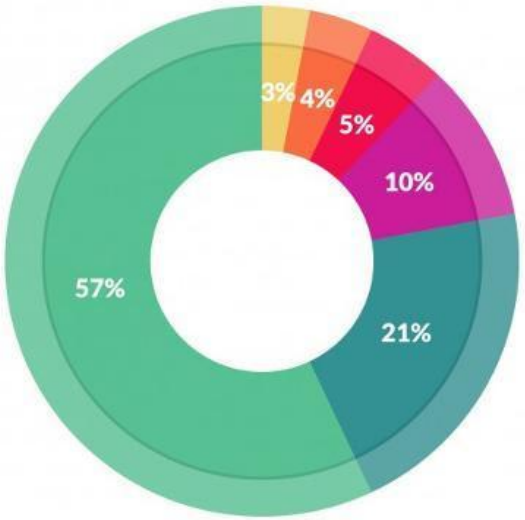


Cleaning and organizing data:

60%

http://www.forbes.com/sites/gilpress/2016/03/23/data-preparation-most-time-consuming-least-enjoyable-data-science-task-survey-says/#f37c7f758459

Key functions of data scientist



What's the least enjoyable part of data science?

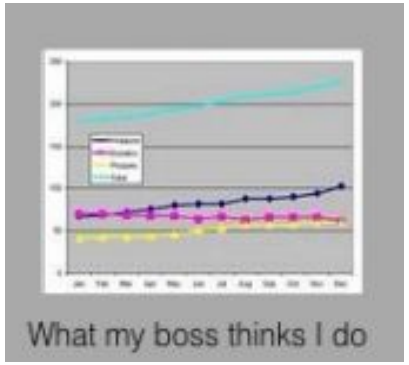
- Building training sets: 10%
- Cleaning and organizing data: 57%
- Collecting data sets: 21%
- Mining data for patterns: 3%
- Refining algorithms: 4%
- Other: 5%

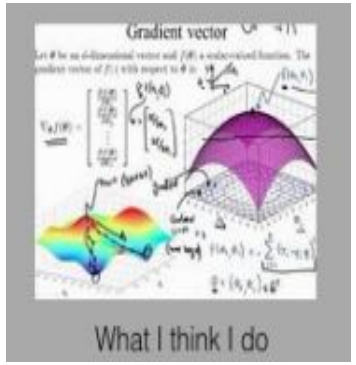
Cleaning and organizing data:

57%

http://www.forbes.com/sites/gilpress/2016/03/23/data-preparation-most-time-consuming-least-enjoyable-data-science-task-survey-says/#f37c7f758459

Key functions of data scientist



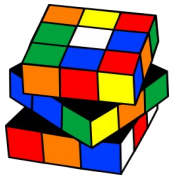


Characteristics of data scientist- YOU MUST LOVE DATA

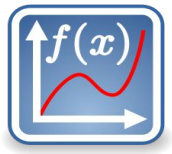
- Common sense
- Curious mind
- Clear and simplify thought

- Good listening, writing and communication skills (visualizations)

- Love to solve puzzles

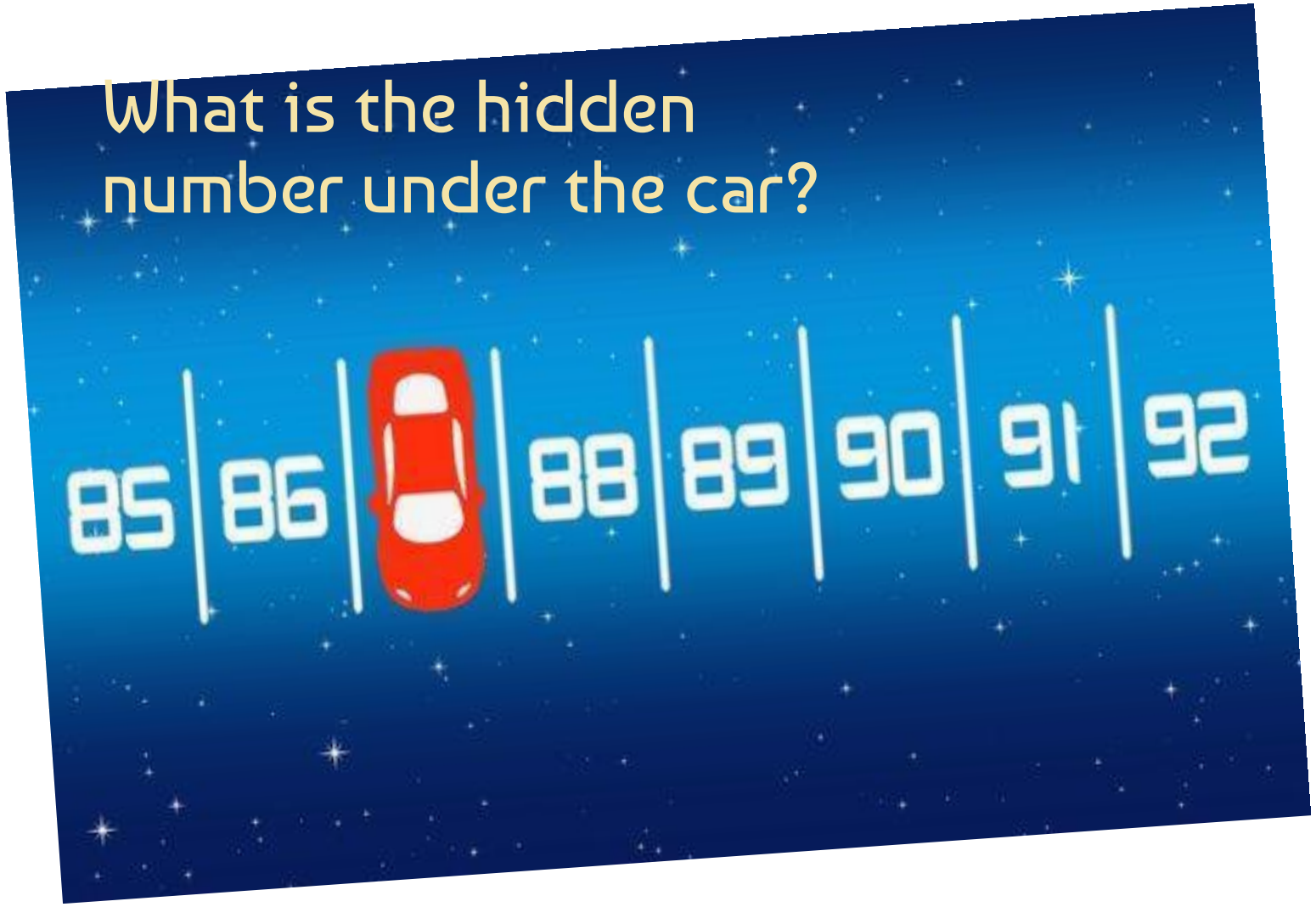


- Maths & Stats



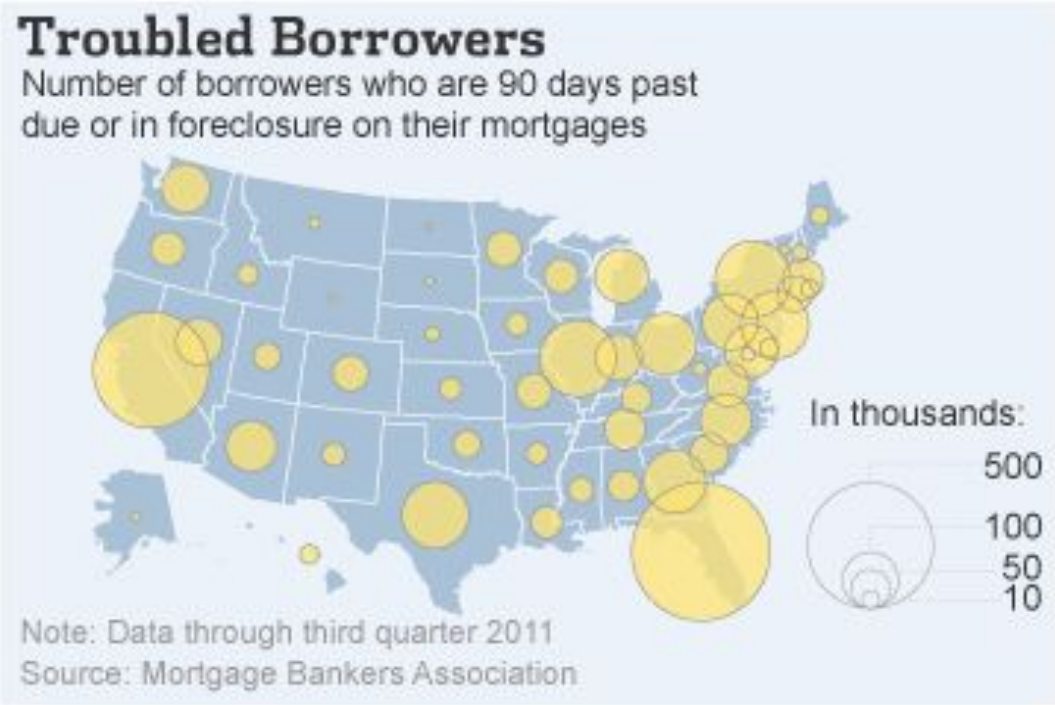
- Business sense

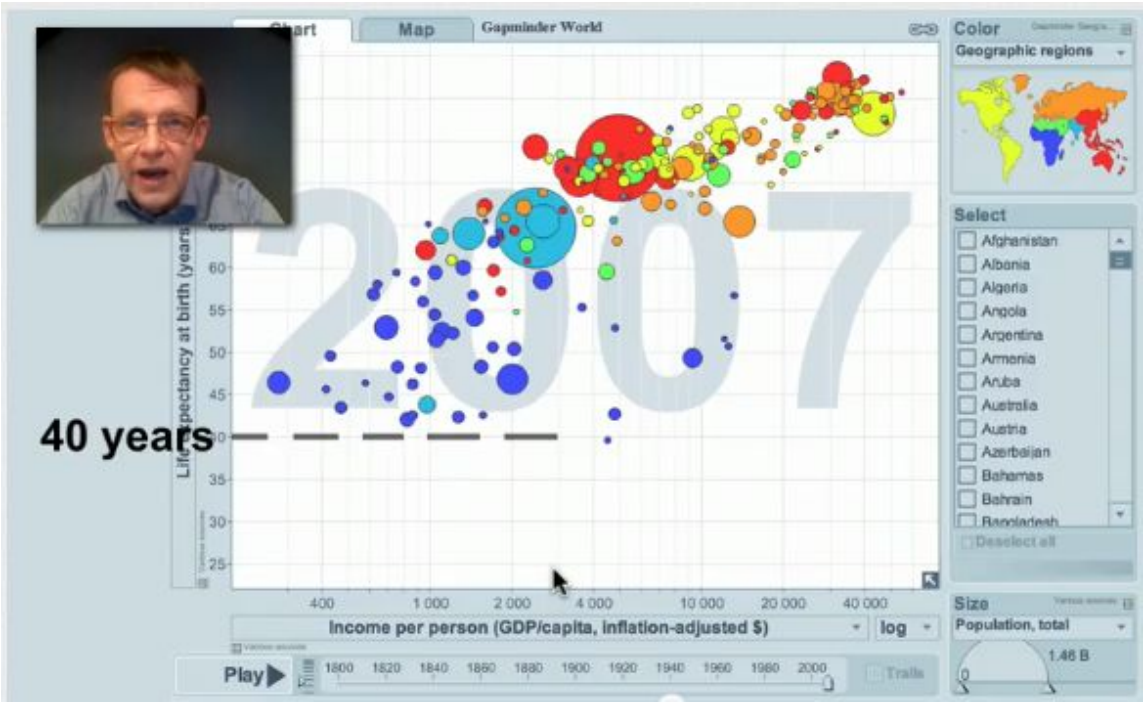
IQ Test



Answer: 87

Data Analytics Applications





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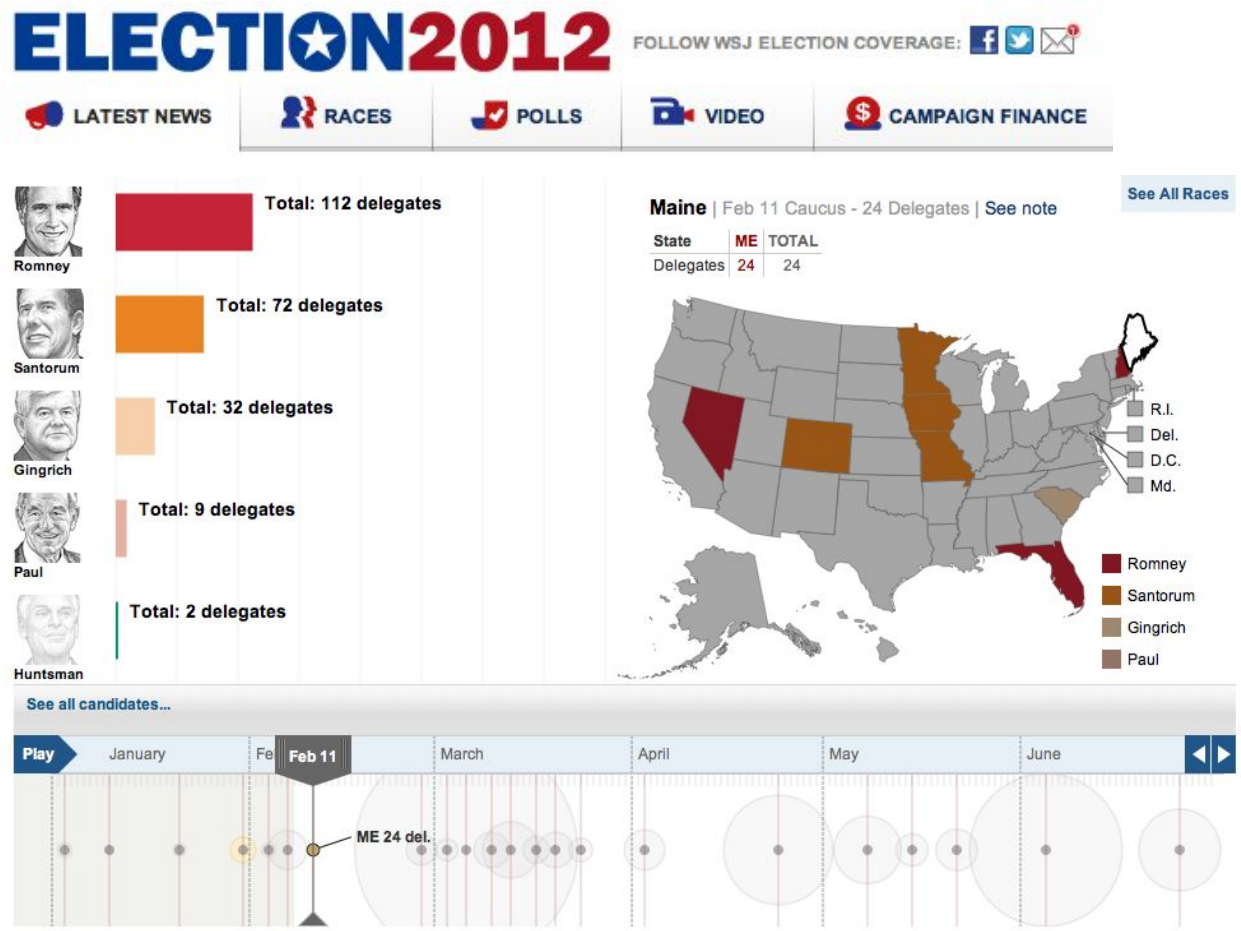

Gartner report on data-driven decision making is critical to create business value

https://www.gartner.com/en/information-technology/insights/data-analytics

Summary

- What did I learn in this module?
  - » State data analytics definition and its importance.
  - » Identify and explain the four types of data analytics.
  - » Identify the role of data scientists.

Key Terms

- **Data**
  - » Numerical or textual facts and figures that are collected through some type of measurement process.
- **Information**
  - » Meaning from data to support evaluation and decision making.

- **Insights, Analysis**

  >> Finding pattern from data to support evaluation and decision making.

References

- Runkler, T. A. (2020). Data analytics. Wiesbaden: Springer Fachmedien Wiesbaden.